

Vegard Pettersen

Portfolio | GitHub | vegardbp@gmail.com | Norway

PROFILE

- Gameplay programmer with a commercial title in development and shipped projects in C++ and Unity.
- BE in Automation & Intelligent Systems, NTNU 2026, average grade **4.43/5**.
- GPU programming in both real-time graphics and general-purpose compute (HLSL/GLSL).
- Built and shipped Steam P2P networking, custom lobby systems, and proximity voice chat.

TECHNICAL SKILLS

Languages: C++, C#, C, Python, HLSL, GLSL

Engines & Tools: Unity, Git, Blender, VCPkg, CLion, Visual Studio, Linux

Specialties: Multiplayer Networking, Compute Shaders, Shaders, Physics, Optimization

EDUCATION

Bachelor of Engineering, Automation

NTNU, 2023–2026

Average Grade: **4.43 / 5.0** | A grades in: Mathematics, Physics, OOP, DSP, Intelligent Systems, and Control Theory.

SELECTED PROJECTS

Dark Bros | *Solo Developer*

Releasing Q3 2026

Tech: Unity, C#, HLSL, Netcode for GameObjects, Steamworks

- A commercial FFA arena brawler on Steam built entirely solo over 1.5 years, covering gameplay code, shaders, 3D models, animations, UI, and sound design.
- Networking built on Steam P2P via Facepunch Steamworks, with a full lobby system (hosting, search filters, kick/ban), proximity voice chat with per-player mute, and a networked grappling hook.
- GPU-instanced grass with compute shader frustum culling; HLSL shaders for volumetric atmosphere, world-space shadows, and subsurface scattering.
- Snapshot-based replay system for full post-match playback of match state.
- Custom animation system built to bypass Unity's default pipeline for faster iteration; modular attack system that abstracts away networking for most player actions.

Roguelike Asteroid Game 2.5D | *Individual Project*

2023

Tech: C++, threepp, VCPkg

- C++ roguelike built from scratch with a custom 3D physics engine and particle system.
- Graded **A**; noted for memory management and clean OOP architecture.

Compute Shader Genetic Algorithm

Exam Project

Tech: Unity, C#, HLSL, Git

- GPU-accelerated genetic algorithm solving the Rastrigin function in high-dimensional search spaces, with the same architecture applicable to problems like multi-dimensional PID tuning.
- Graded **A**; noted for algorithm design and documentation.

WORK EXPERIENCE

Staff | *Buldrehallen, Norway*

2025 (6 months)

- Ran solo shifts of 5–8 hours with full responsibility for safety, security, and customer experience.
- Handled POS transactions, membership registrations, and food and beverage service.
- Gave safety briefings and technical instruction to new climbers.
- Reference available upon request.